# Dairy Farm Performance Dashboard

# Power BI Analytics for Dairy Supply Chain & Sales

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## 1. Problem Statement

The dairy industry generates vast amounts of data across production, inventory, sales, and logistics. However, dairy farm managers and supply chain analysts often struggle with:

- **Fragmented data** – Information is scattered across spreadsheets, making it difficult to see the big picture.
- **Inventory mismanagement** – Without real-time stock monitoring, farms face stockouts or excess wastage of perishable products.
- **Sales channel ambiguity** – Unclear which sales channels (retail, wholesale, online) drive the most revenue.
- **Geographic blind spots** – No visual way to identify high-revenue vs. underperforming customer locations.
- **Farm efficiency opacity** – No insight into how land area, number of cows, and farm size correlate with production.

A unified, interactive dashboard is needed to transform raw dairy data into **actionable business intelligence**.

## 2. Proposed Solution

This project delivers a **four-page interactive Power BI dashboard** built on a comprehensive dairy dataset containing over 2,000 records. The dashboard empowers stakeholders to:

- Monitor **key performance indicators (KPIs)** at a glance.
- Track **inventory levels** and identify products at risk of stockout.
- Analyse **sales performance** across channels, products, brands, and locations.
- Visualise **farm productivity** based on land area, herd size, and farm category.
- Make **data-driven decisions** regarding production planning, reorder quantities, and channel focus.

## 3. Dashboard Features & Pages

The dashboard is structured into four focused pages, each addressing a distinct business area.

### Page 1: Executive Summary

- **KPIs**: Total Revenue, Quantity Sold, Profit Margin %, Average Price.
- **Trend Analysis**: Monthly revenue and quantity sold trend lines.
- **Top Performers**: Horizontal bar chart of top products by revenue.
- **Brand Share**: Donut chart visualising revenue distribution by brand.
- **Geographic Map**: Revenue bubbles mapped across Indian states.

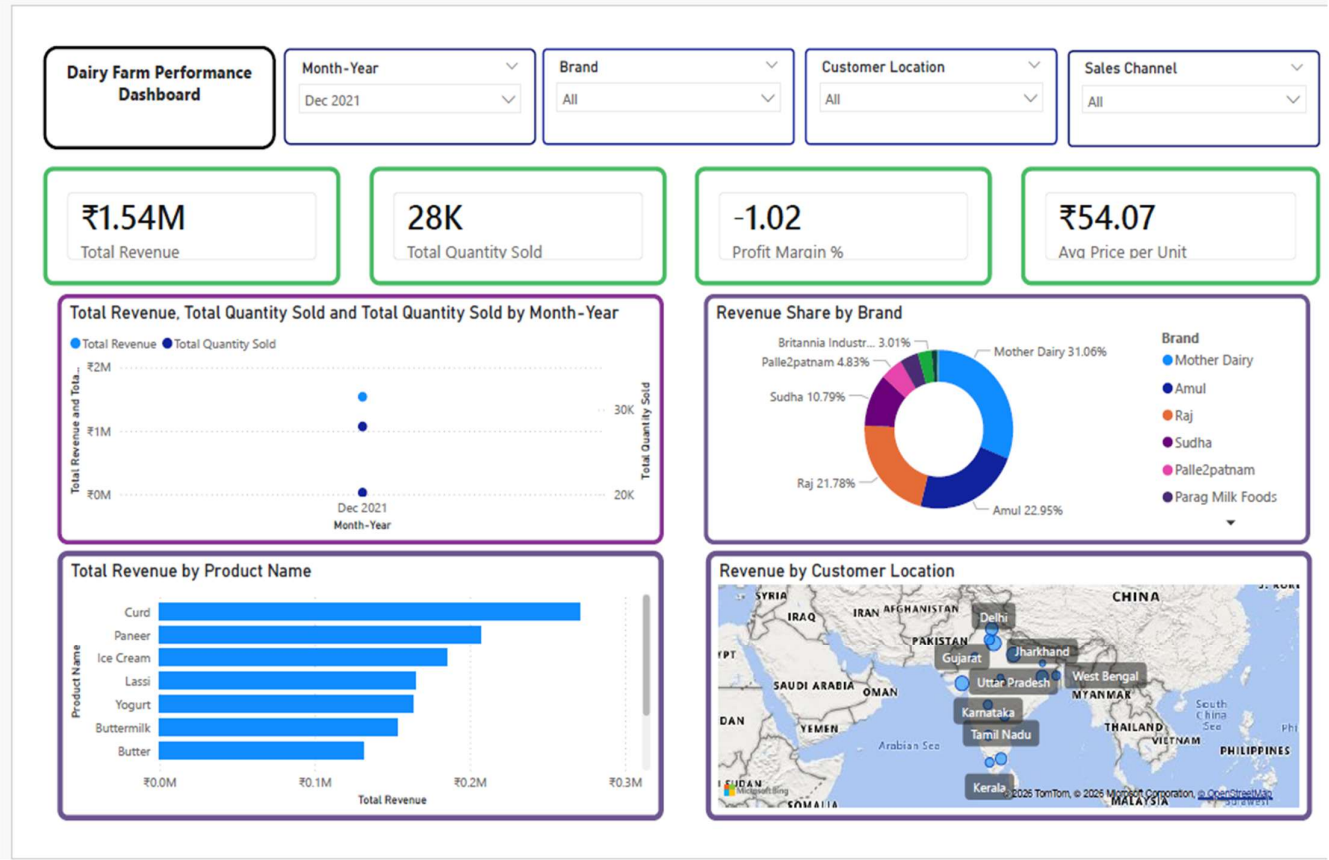


*Figure 1: Executive Summary Page*

## Page 2: Inventory & Operations

- **Stock Health**: Card showing Stockout Risk % and Average Shelf Life.
- **Stock vs Threshold**: Stacked bar chart comparing current stock with minimum required levels.
- **Stock Status Table**: Colour-coded table highlighting products "Below Threshold" in red.
- **Shelf Life vs Revenue**: Scatter plot analysing relationship between product shelf life and revenue.

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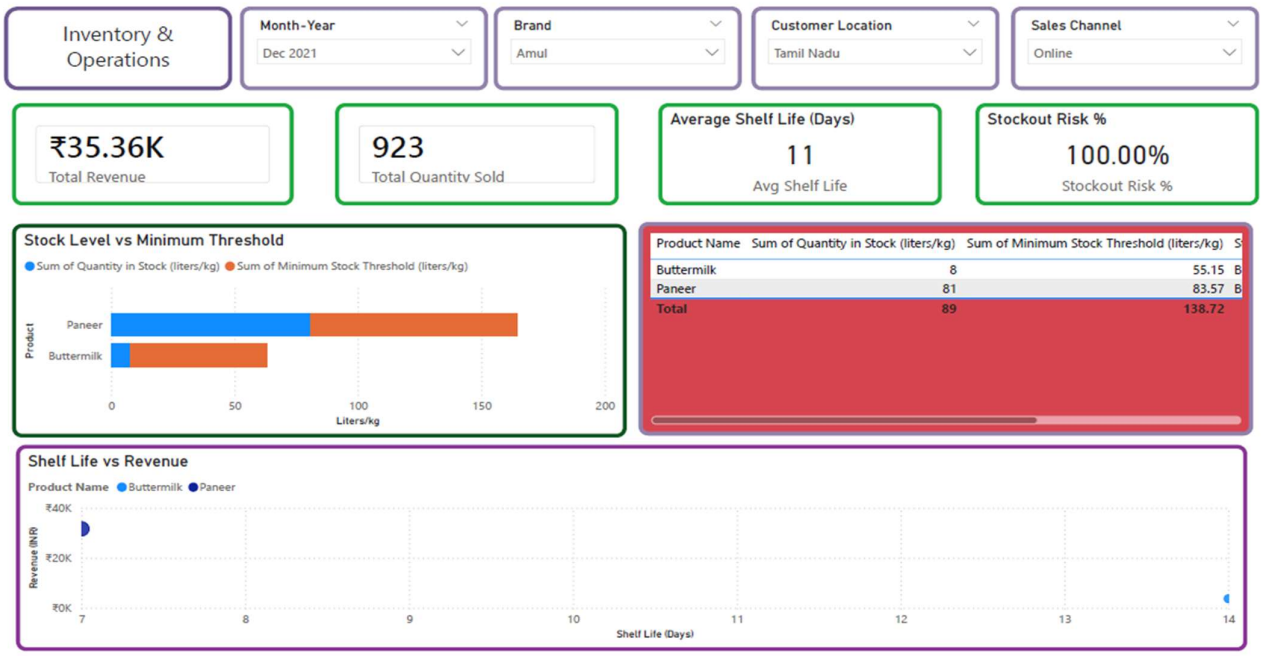


*Figure 2: Inventory & Operations Page*

## Page 3: Sales & Channel Analysis

- **Channel & Product**: Clustered column chart of revenue by sales channel (Retail, Wholesale, Online) split by product.
- **Geographic Treemap**: Rectangles sized by revenue per customer location.
- **Monthly Channel Trend**: Line chart showing revenue over time for each sales channel.
- **Product-Brand Matrix**: Cross-tabulation of revenue and quantity sold by product and brand.

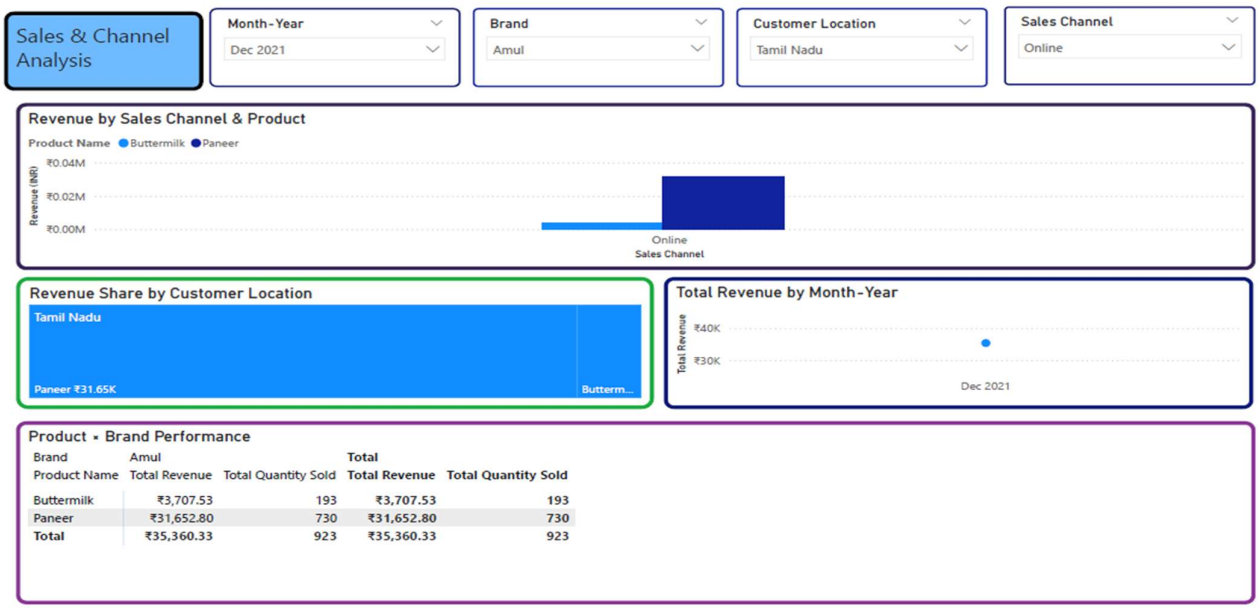


*Figure 3: Sales & Channel Analysis*

## Page 4: Farm & Production Insights

- **Land vs Cows Scatter**: Bubble chart showing farms by land area and cow count; bubble size = production volume.
- **Farm Size Productivity**: Bar chart comparing average production quantity across small, medium, and large farms.
- **Farm Details Table**: Tabular view of each farm's location, size, land area, cows, and total production.

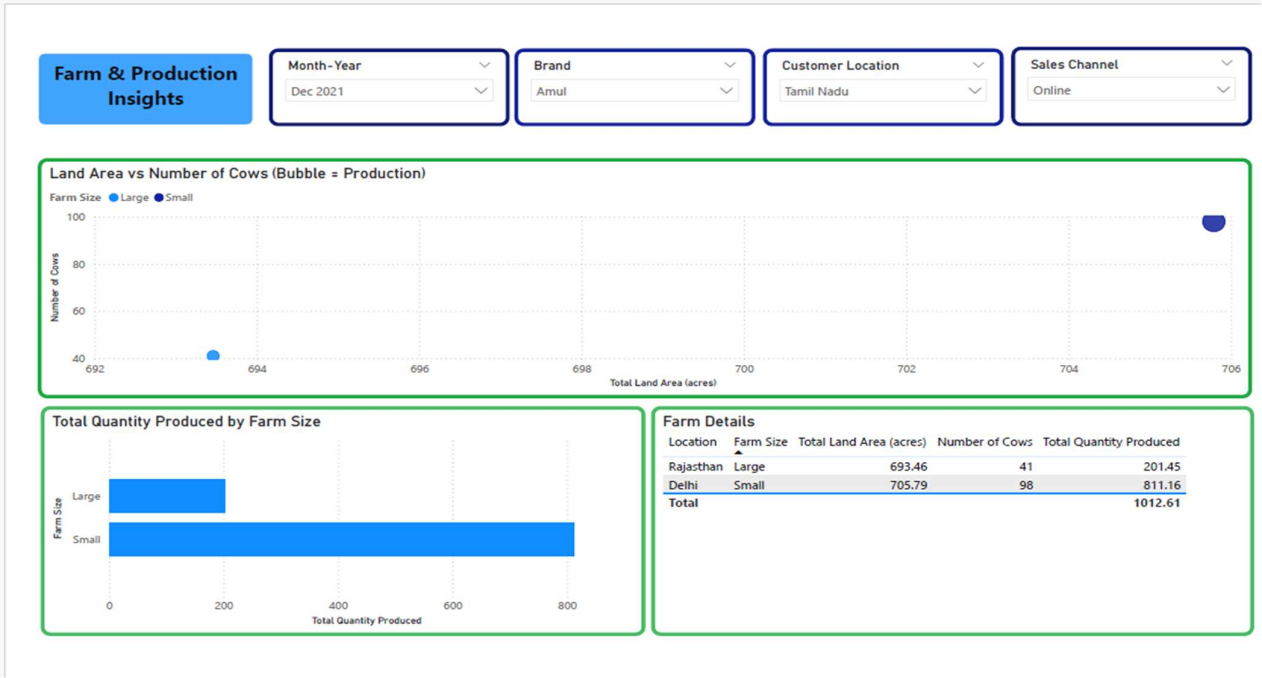


*Figure 4: Production Insights*

## 5. Technology Stack

| Component | Technology Used |
|---|---|
| Data Source | CSV file (dairy_dataset.csv) |
| Data Transformation | Power Query (M language) |
| Data Modelling | DAX (Data Analysis Expressions) |
| Visualisation | Microsoft Power BI Desktop |
| Documentation | Microsoft Word / PDF |

### Key DAX Measures Created:

- Total Revenue, Total Quantity Sold, Total Quantity Produced
- Profit Margin %, Stockout Risk %, Avg Shelf Life
- Inventory Turnover, Avg Price per Unit

### Power Query Transformations:

- Data type correction (dates, decimals, text)
- Text trimming and conditional column for Farm Size Sort
- Creation of Month-Year and Month-Year Sort columns for time intelligence

## 6. Unique Points / Highlights

1. **End-to-End Analytics**: Covers production, inventory, sales, and farm operations in one unified solution.
2. **Real-Time Inventory Alerts**: Conditional formatting instantly flags products needing reorder.
3. **Custom DAX Metrics**: Measures like *Stockout Risk %* and *Inventory Turnover* provide operational insights beyond standard aggregations.
4. **Geospatial Visualisation**: Interactive map pinpoints revenue distribution across Indian states.
5. **Scalable Design**: The data model can easily accommodate additional years, products, or locations.

## 7. Future Improvements

| Improvement | Description |
|---|---|
| **Automated Data Refresh** | Connect to a live SQL database or cloud source for real-time updates. |
| **Predictive Analytics** | Integrate AI visuals (e.g., decomposition tree, key influencers) to forecast sales and stock requirements. |
| **Row-Level Security (RLS)** | Restrict data access based on user roles (e.g., regional managers see only their state). |
| **Mobile-Optimised Layout** | Design a phone-friendly view for on-the-go monitoring. |
| **What-If Parameterisation** | Allow users to simulate changes in price or reorder quantity and see impact on revenue/margins. |